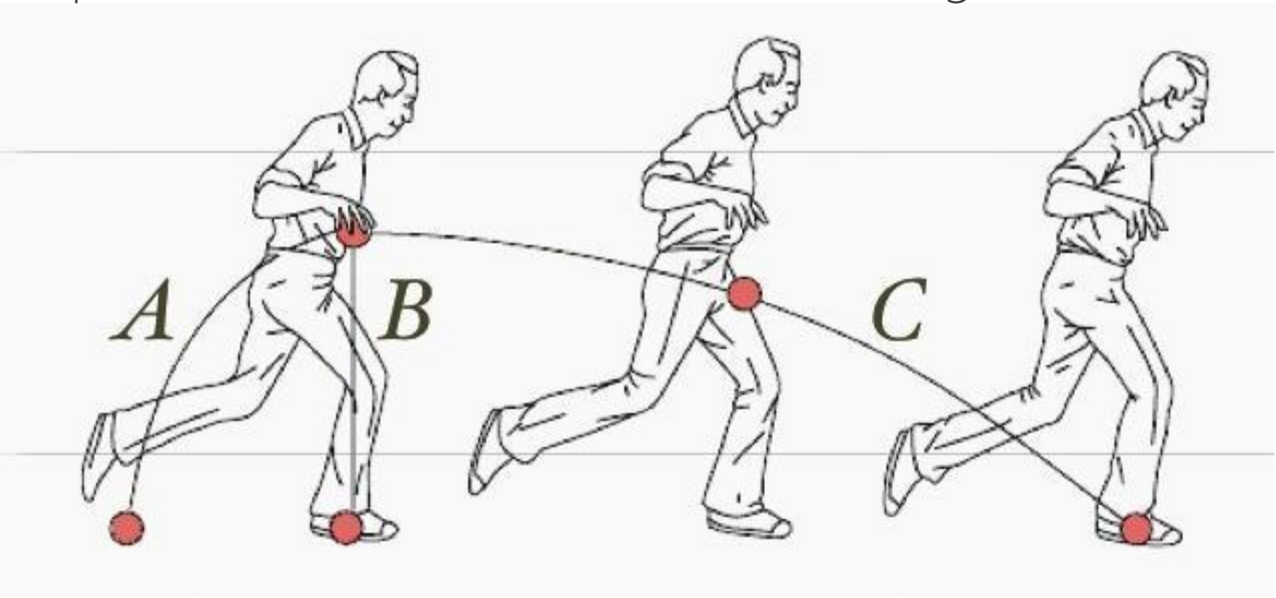
Mental models are important!

People are explanatory creatures, they can’t help to build mental models to help form explanations of things—even when these explanations are wrong.

[Norman, The Design of Everyday Things, 1988]

MentalModel: Physics

Which path does the ball take as it falls to the ground: A, B, or C?



When asked of sixth graders, **only 3 percent answered C**, the right answer.

When asked of high school students who had studied six weeks of Netownian mechanics, **only 20 percent answered C**.

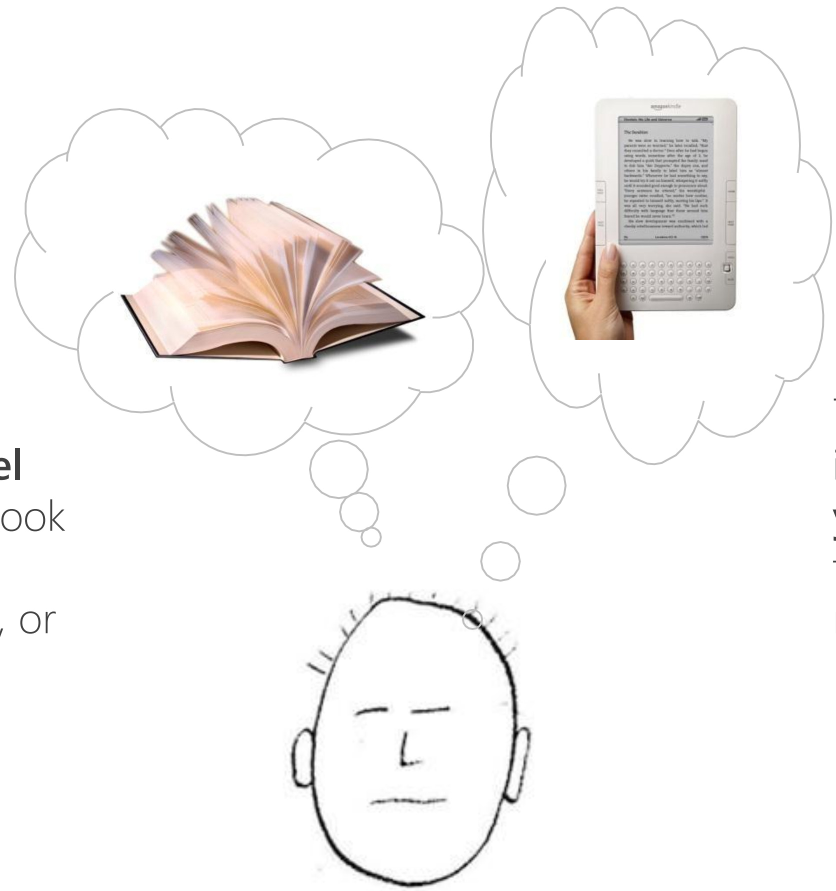
[Example from The Design of Everyday Things by Don Norman; Original study White & Horwitz, 1987]

What do you do with this?

How about now?

Imagine that you've never seen an iPad. I hand you one and tell you that you can read books with it.

You "automatically" start **building a mental model** about how the book will look and act on the screen—things like turning a page, or inserting a bookmark



You

This **mental model** will be **influenced by**, for example, **your experience** with touchscreens or other ebook readers.

[Weinschenk, The Secret to Designing an Intuitive UX : Match the Mental Model to the Conceptual Model, UX Magazine, 2010]

Imagine that you've never seen an iPad. I hand you one and tell you that you can read books with it.

You

This **mental model will change** as you interact with the iPad and read a few books.

[Weinschenk, The Secret to Designing an Intuitive UX : Match the Mental Model to the Conceptual Model, UX Magazine, 2010]

A **mental model** represents a person's thought process for **how something works** (i.e., a person's understanding of the surrounding world).

Susan Carey
Harvard Professor in Psychology

[Carey, Cognitive Science and Science Education, 1986]

A **mental model** represents a person's thought process for **how something works** (i.e., a person's understanding of the surrounding world). Mental models are **based on incomplete facts, past experiences, and even intuitive perceptions**.

SusanCarey

Harvard Professor in Psychology

[Carey, Cognitive Science and Science Education, 1986]

A **mental model** represents a person's thought process for **how something works** (i.e., a person's understanding of the surrounding world). Mental models are **based on incomplete facts, past experiences, and even intuitive perceptions**. They help **shape actions and behavior**, influence what people pay attention to in complicated situations, and define how people approach and solve problems.

SusanCarey
Harvard Professor in Psychology

[Carey, Cognitive Science and Science Education, 1986]

Mental models

"In interacting with the environment, with others, and with the artifacts of technology, **people form internal, mental models of themselves and of the things with which they are interacting.**

These models provide **predictive** and **explanatory** power for understanding the interaction."

– Norman (in Gentner & Stevens, 1983)

OK, so **users** have **mental models**, not a big surprise…

Designers also have **mental models** both for how their design should operate and how users might perceive their design!

Mental models vs. Conceptual Design

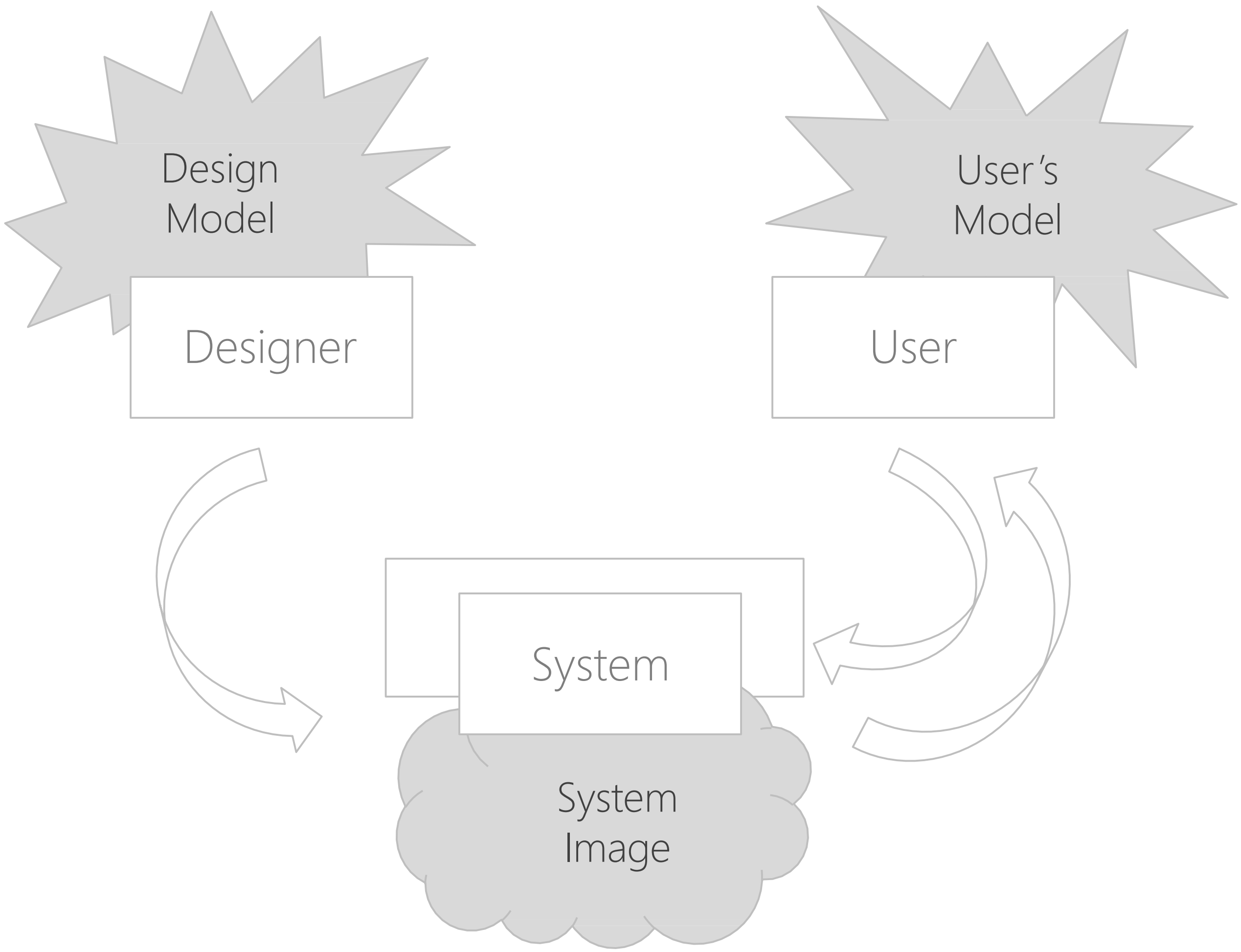
Mental models: *something the **user has (forms)***

- Users "**see**" the system through mental models
- Users **rely** on mental models during usage
- There are various **forms** of mental models
- Mental models can **support** users‟ interaction

Conceptual design: *something the **designer does***

- **Defining** the *intended* mental model
  - Hiding the technology of the system
- **Designing** a suitable system image
  - Applying appropriate design guidelines
- **Analysis** using "walkthroughs"

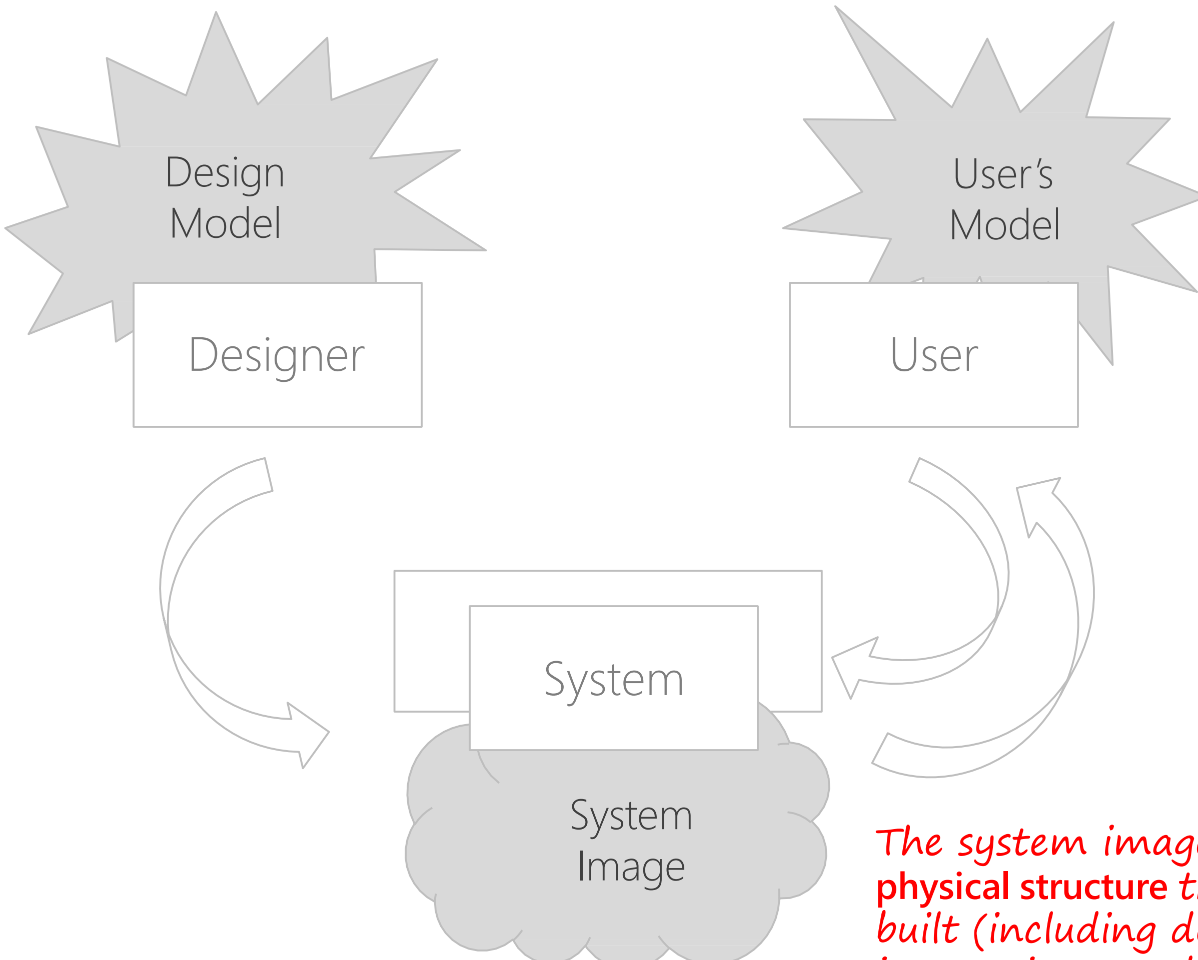
Conceptual Model



[Norman, The Design of Everyday Things, 1988]

ConceptualModel

The design model is the **designer's conceptual model**

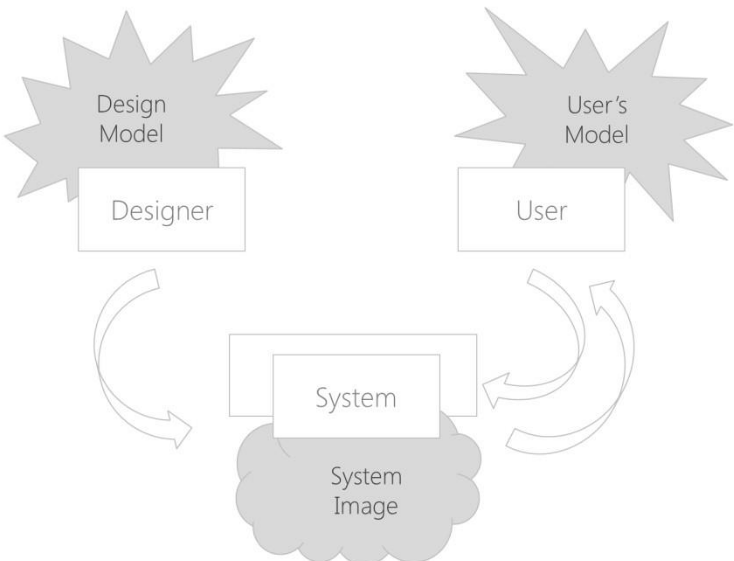


The user's model is the **mental model** developed through interaction with the system (and past experiences)

The system image **results from the physical structure** that has been built (including documentation, instructions, and labels)

[Norman, The Design of Everyday Things, 1988]

TheProblem



The designer expects the **user's model to be identical** to the **design model**. It rarely is!

The designer communicates to the user through the design.

If the **system image** does not make the **design model clear and consistent**, the user will end up with the **wrong mental model**!

[Norman, The Design of Everyday Things, 1988]

Various models

- **Design model** is the designer‟s conceptual model
- **System model** is a model of the way the system works
- **System image** results from the physical structure of what has been built (including documentation, instructions, labels) – it is what the user "sees"
- **User's model** is the "**mental model"** developed by the user through interaction with the system
  - User tries to match the mental model to the system model

Conceptual mismatch

- **Misconceptions** happen when user‟s model differs from the system model

Some characteristics of mental models

- Incomplete
- Constantly evolving
- Not accurate representation
  – (contain errors and uncertainty measures)
- Provide a simple representation of a complex phenomena
- Can be represented by a set of if-then-else rules

Acquiring mental models

- **During system usage**:
  - The user‟s own activity leads to a mental model
  - Explanatory theory, developed by the user
  - Often used to predict future behavior of the system
- **Observing others** using the system:
  - Casual observation of others working
  - Asking someone else to “do this for me”
  - Formal training sessions
- **Reading** about a system
  - Documentation, help screens, “for Dummies” books

This is done **by the user** (not the designer)

Please pour me some hot tea.

Carelman's "Coffeepot for Masochists"

[Example from The Design of Everyday Things by Don Norman]

Let’s go for a bike ride.

Carelman’s “Convergent Bicycle (Model for Fiancés)”

[Example from The Design of Everyday Things by Don Norman]

Let's go for a bike ride.

Why not?

Carelman's "Convergent Bicycle (Model for Fiancés)"

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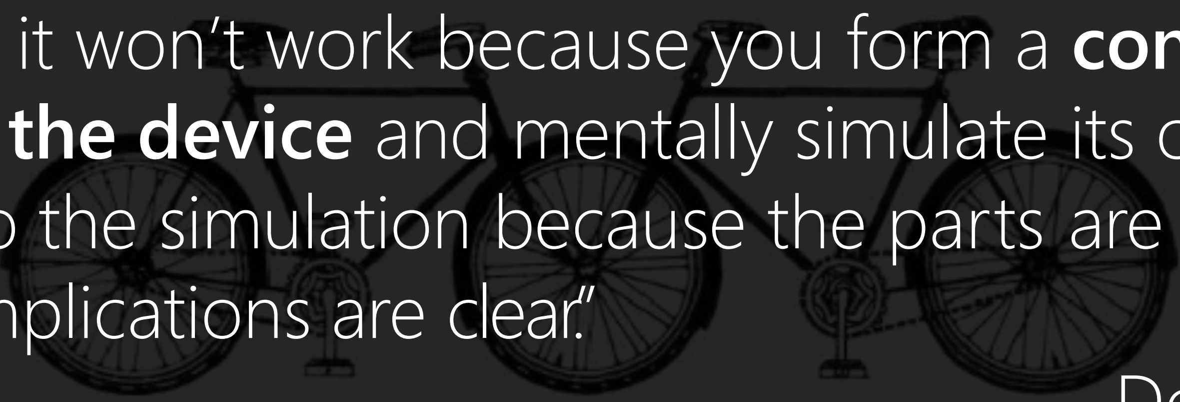
Let's go for a bike ride.

Why not?



"You know it won't work because you form a **conceptual model of the device** and mentally simulate its operation. You can do the simulation because the parts are visible and the implications are clear."

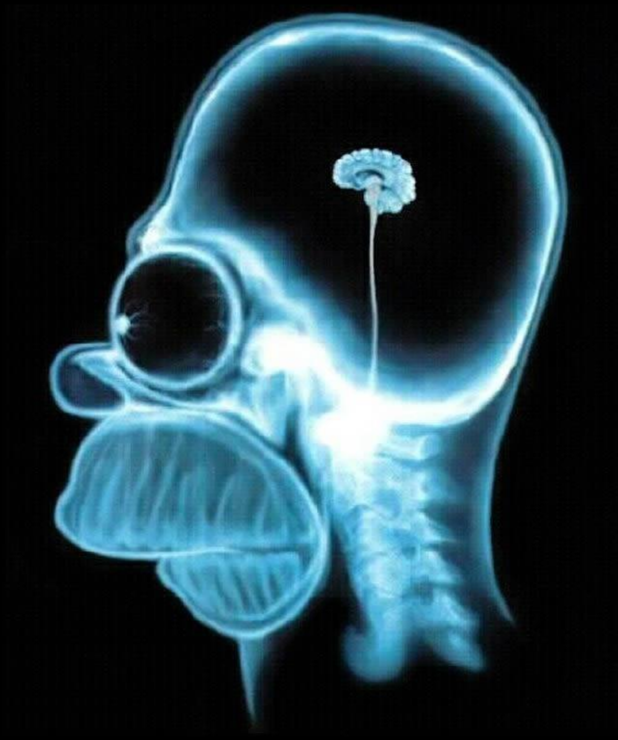
–Don Norman

Carelman's "Convergent Bicycle (Model for Fiancés)"

[Example from The Design of Everyday Things by Don Norman]

A good conceptual model allows us to predict the effects of our actions.

Forming a conceptual model comes from, in part: **affordances**, **constraints**, & **mappings**



Affordances, Constraints, Mappings

[Example from The Design of Everyday Things by Don Norman]

Some Design Principles

Visibility: Can I see what to use? It should be obvious what a control is used for

Affordance: How do I use it? It should be

obvious how a control is used

Feedback: What is it doing now? **:** It should be obvious when a
control has been used

Mapping: How does it relate?

Constraint: I can do this, but I can't do that.

Consistency: Have I seen this before?

Affordances, Constraints, Mappings

Size of holes **constrain** size and number of fingers

Holes **afford** putting something through them. Obvious something=fingers!

Mapping between finger, holes, and blade is **constrained** by design

[Example from The Design of Everyday Things by Don Norman]

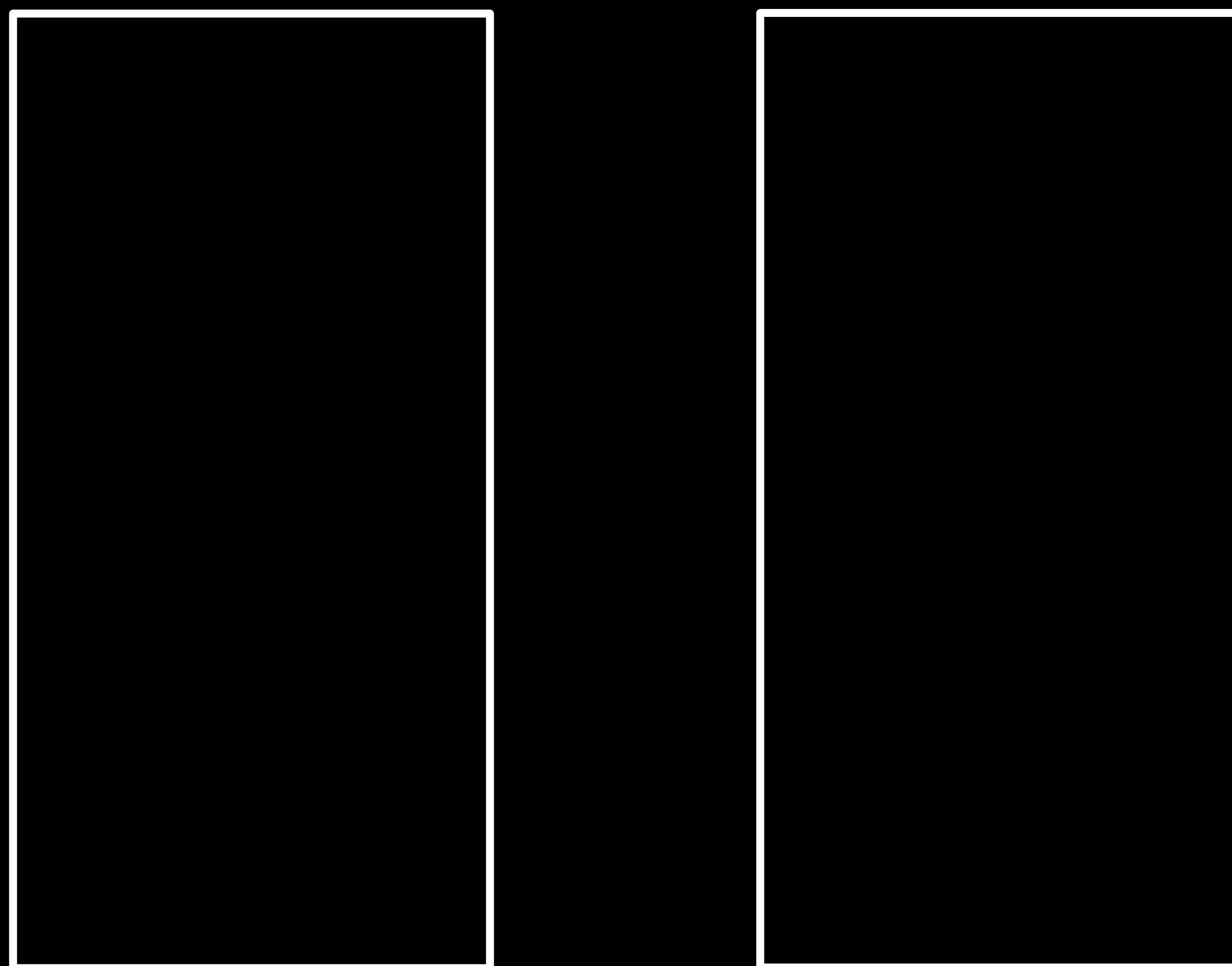
How do you set the time?

It's difficult to tell. There is **no evident relationship** between the operating controls and the functions, no constraints, no apparent mappings.

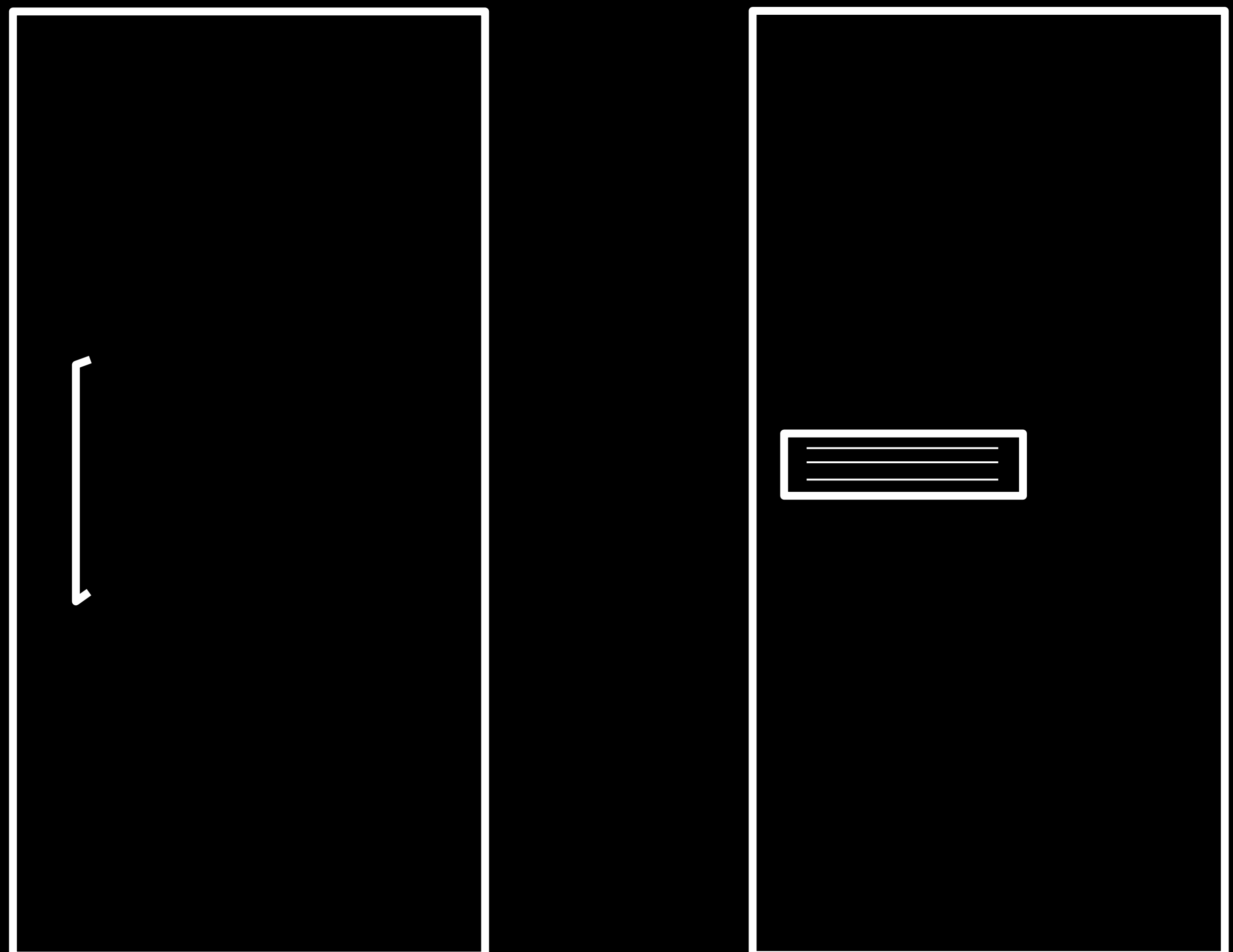
[Example from The Design of Everyday Things by Don Norman]

Affordances

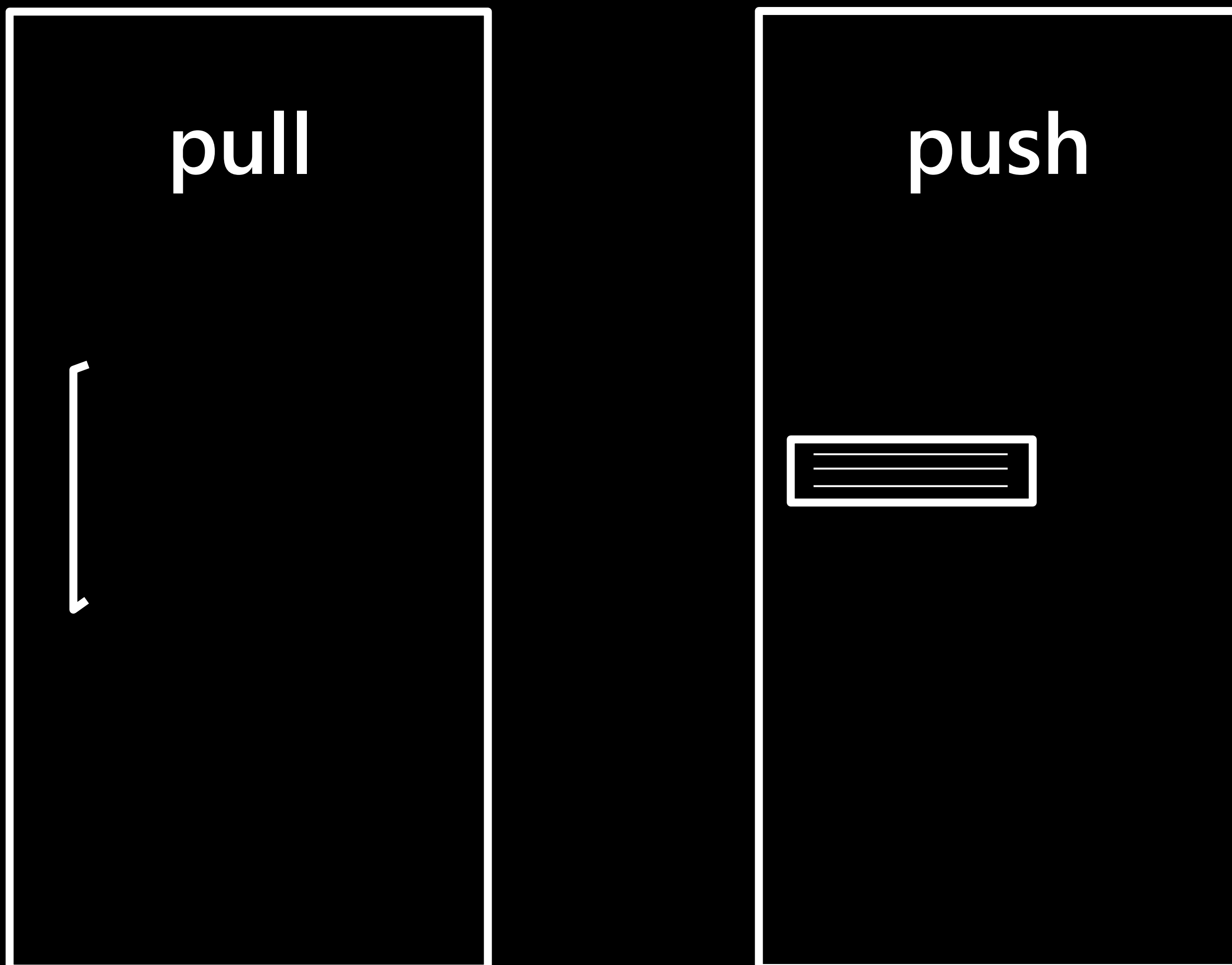
Which door do I **push?**
Which door do I **pull?**



Which door do I **push?**
Which door do I **pull?**



Which door do I **push?**
Which door do I **pull?**



Which door do I **push,**
Which door do I **pull?**


PULL


PUSH

Norman door!

Affordance refers to the perceived & actual properties of a thing that determine just how that thing could possibly be used

Don Norman
Cognitive Scientist / Author

vertical handle cues pull behavior


PUSH


PUSH
FAIL #1


PULL
PULL

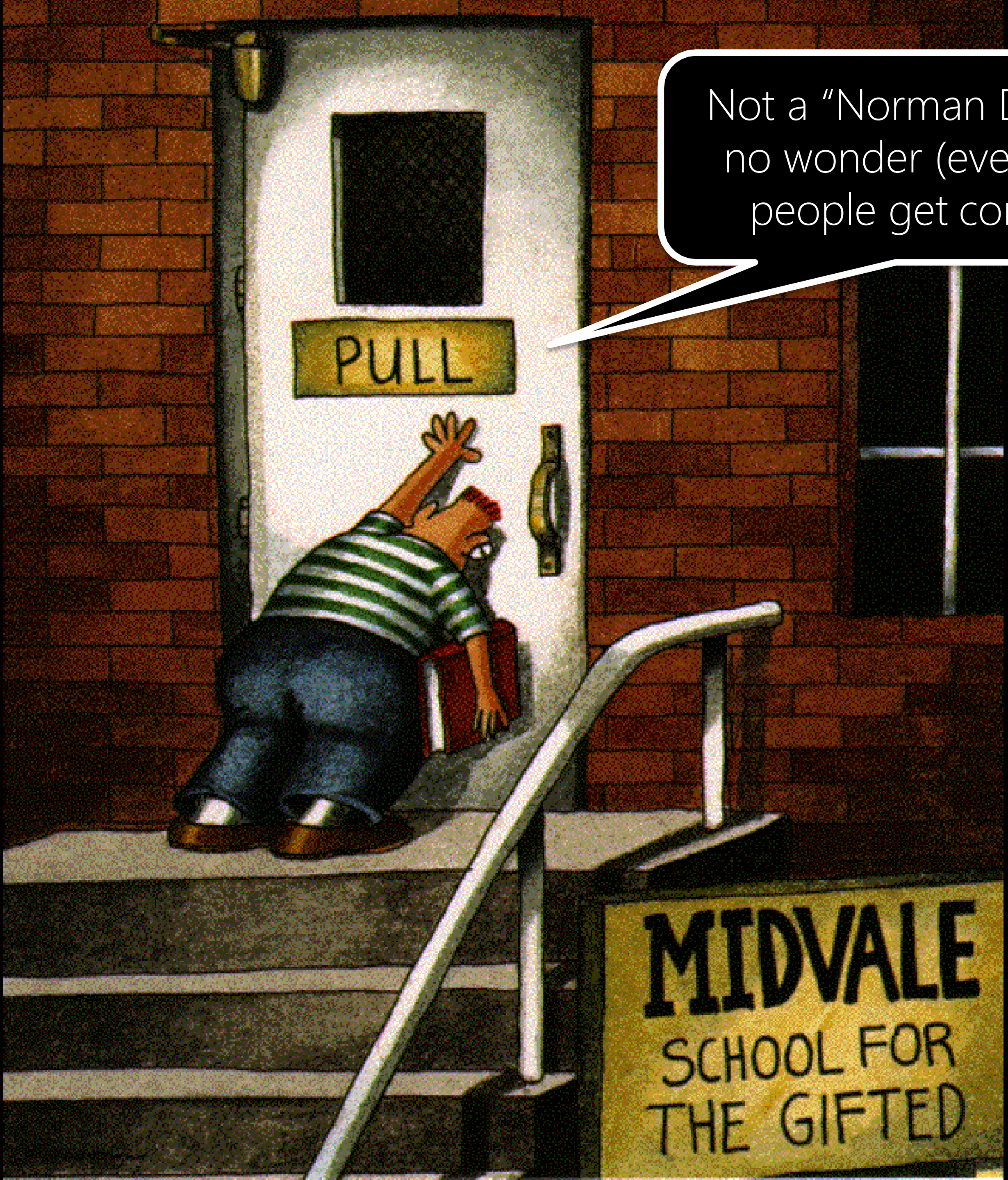

PULL
PULL
FAIL #2

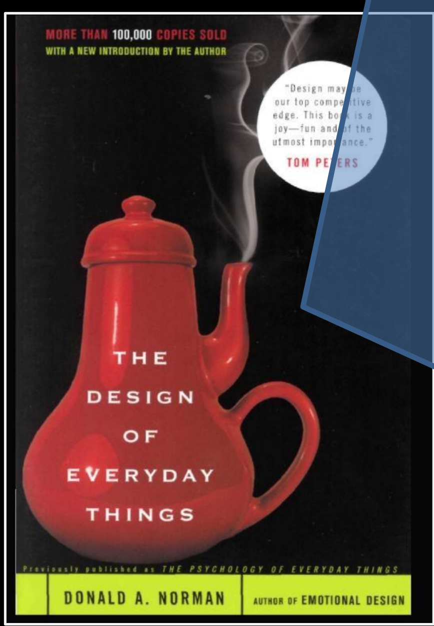

DUNK IT IN
DROP IT IN
PUSH
You can HAVE IT YOUR WAY™ and pull it
you want, but this hinge is pretty stubborn.
BURGER KING
AMERICAN EXPRESS
Pull
BURGER KING

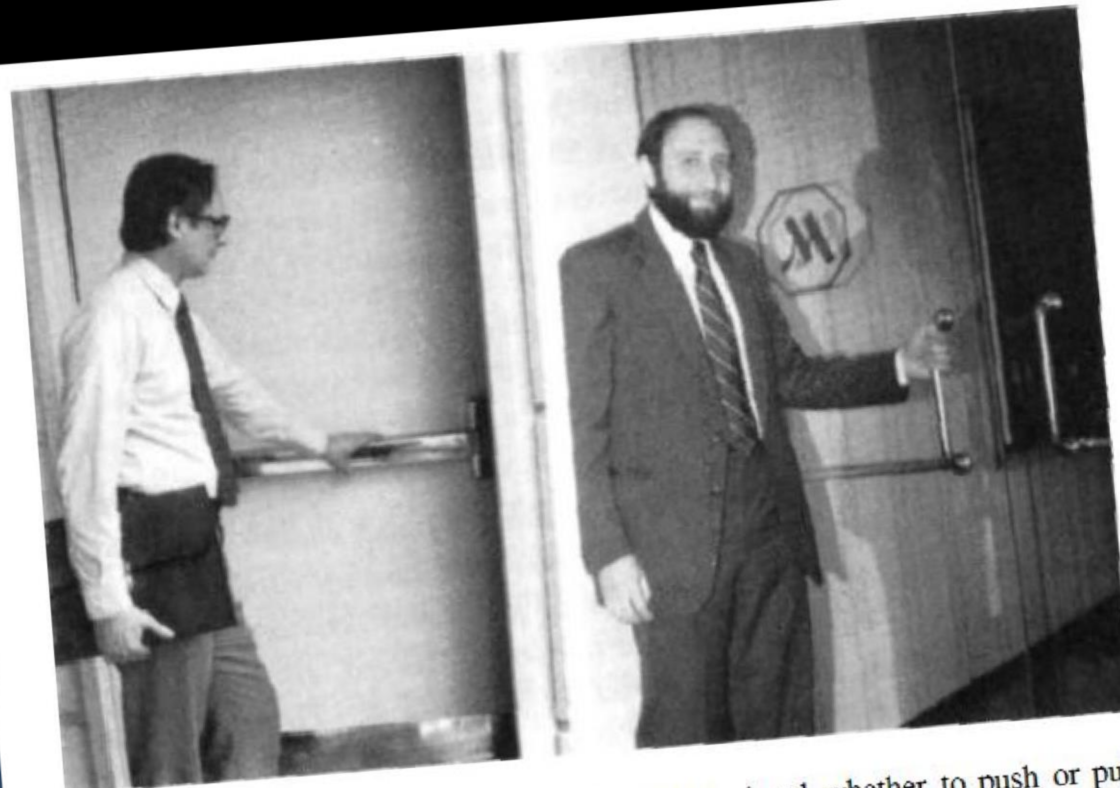

FAIL #3
PUSH
You can HAVE IT YOUR WAY™ and pull if you want, but this hinge is pretty stubborn.
BURGER KING
AMERICAN EXPRESS
Pull
BURGER KING


FAIL #3
PUSH
You can HAVE IT YOUR WAY™ and pull it if you want, but this hinge is pretty stubborn.
BURGER KING
AMERICAN EXPRESS
Pull
FAIL #4
BURGER KING

Not a “Norman Door” but no wonder (even gifted) people get confused!



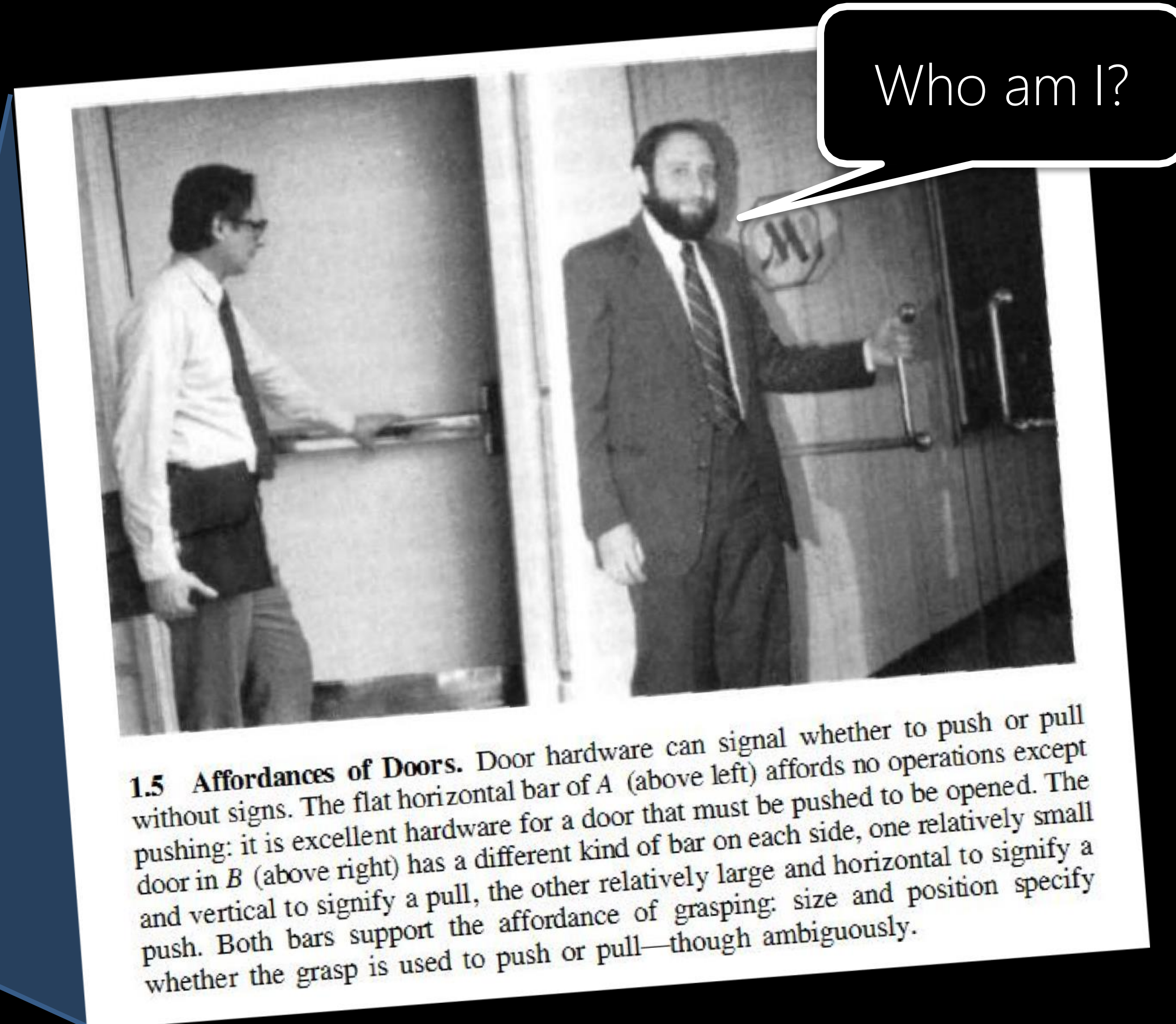




1.5 Affordances of Doors. Door hardware can signal whether to push or pull without signs. The flat horizontal bar of *A* (above left) affords no operations except pushing: it is excellent hardware for a door that must be pushed to be opened. The door in *B* (above right) has a different kind of bar on each side, one relatively small and vertical to signify a pull, the other relatively large and horizontal to signify a push. Both bars support the affordance of grasping: size and position specify whether the grasp is used to push or pull—though ambiguously.

The Design of Everyday Things

by Don Norman



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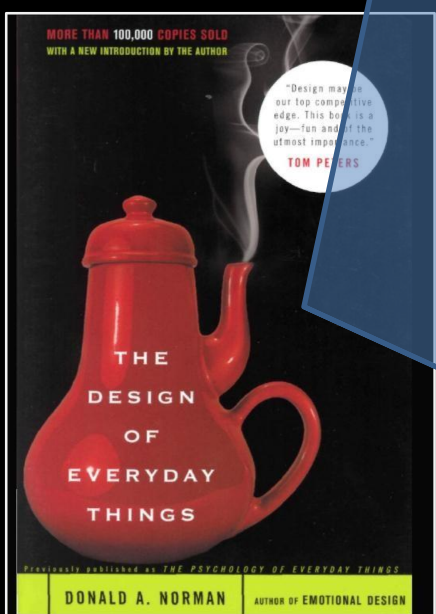


The Design of Everyday Things
by Don Norman

Cultural Dependencies

Affordances suggest how to use the object

Can be dependent on the

Experience

Knowledge

Culture

[From Bjoern Hartmann, Prof. at UC-Berkeley]

Cultural Dependencies

Affordances suggest how to use the object

Can be dependent on the

Experience

Knowledge

Culture

Switches (US down=off, UK down=on)

red = danger, green = go

Can make an action easy/difficult

[From Bjoern Hartmann, Prof. at UC-Berkeley]

Perceived Affordances

Affordances suggest how to use the object

Can be dependent on the
- Experience
- Knowledge
- Culture of the actor

Can make an action easy/difficult

Affordances may be *perceived*
without actually existing

[From Bjoern Hartmann, Prof. at UC-Berkeley]

Affordances affect our perception of use

Can affordances change our **behavior?**


In-N-Out
Burger!


Environmental protection
可回收物
Recyclable
纸类、玻璃、金属、塑料等
政府提倡使用的资源再生利用产品
爱我家园
Love our homeland
其它垃圾
Other waste
剩饭菜、瓜果壳、灰土等
政府提倡使用的资源再生利用产品


WASTE
WASTE
CANS & BOTTLES
CANS & BOTTLES
NEWSPAPERS
NEWSPAPERS

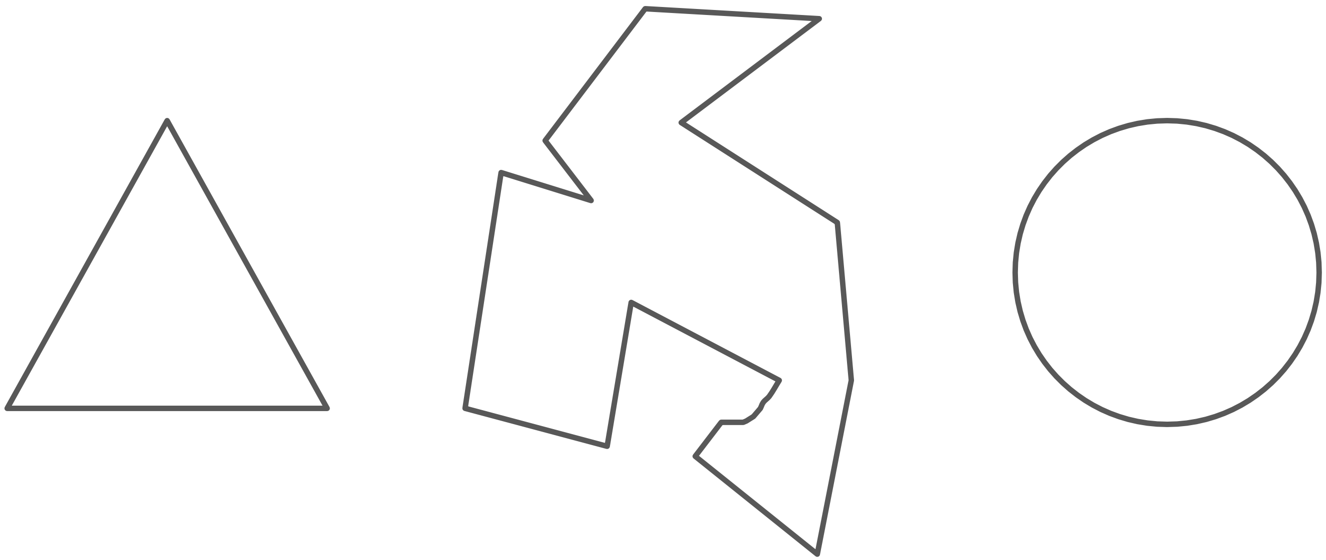
Battle of the Cans


WASTE
WASTE
CANS & BOTTLES
CANS & BOTTLES
NEWSPAPERS
NEWSPAPERS
Winner!
34% increase
in recycling
Duffy, *Environment and Behavior*, 2010


WASTE
WASTE
CANS & BOTTLES
CANS & BOTTLES
NEWSPAPERS
NEWSPAPERS
Holes constrain behavior and also remind what is recyclable
Duffy, *Environment and Behavior*, 2010

Affordances in **UI design**

What are the **affordance** of these graphical objects?



[From Bjoern Hartmann, Prof. at UC-Berkeley]

How about these?

[From Bjoern Hartmann, Prof. at UC-Berkeley]

I’m a button!

I’m a button!

I’m a button!

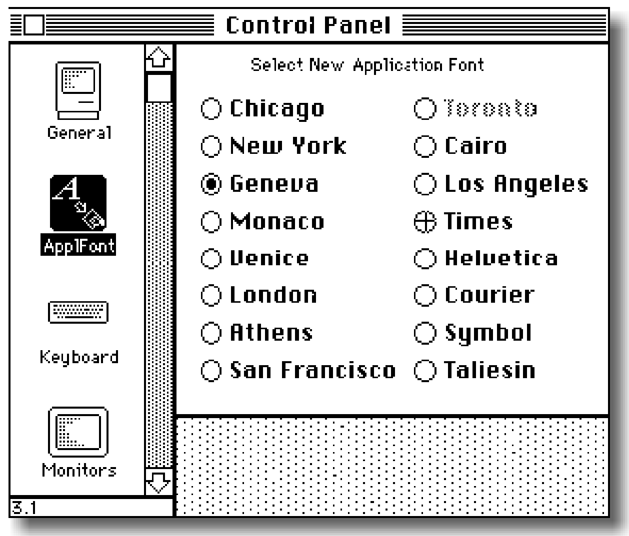
I’m a button

I’m a button

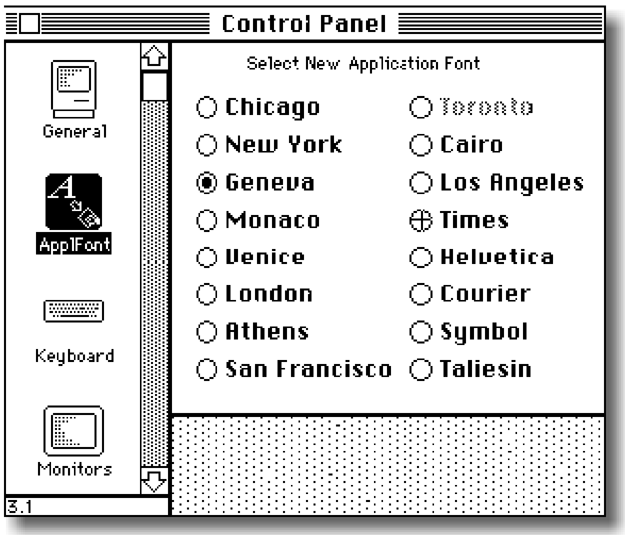
I’m a button

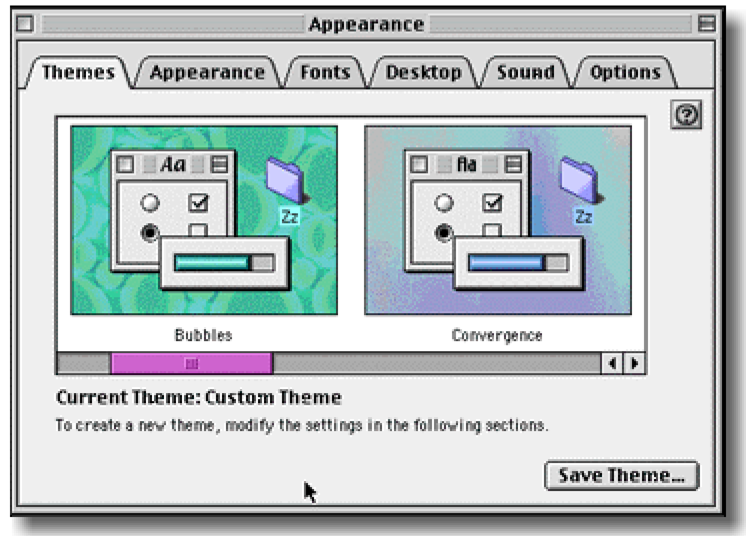
Graphic design emphasizes affordances
Helps user recognize objects as buttons

What does a click on the left pane do?

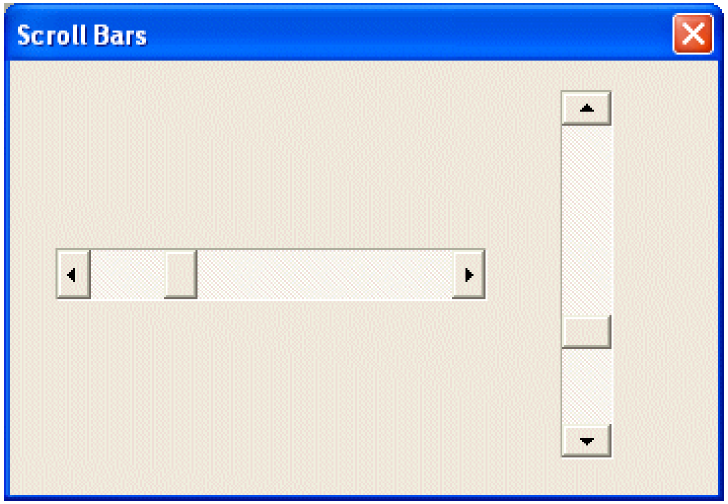


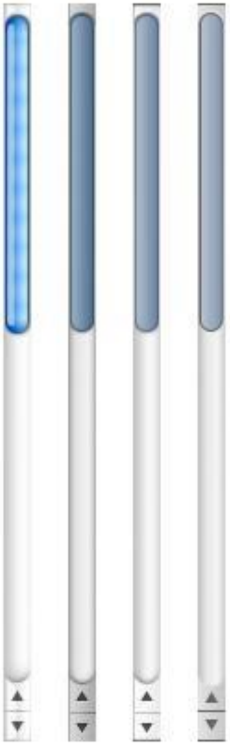
What does a click on the left pane do?





What are the **scrollbar affordances?**





[From Bjoern Hartmann, Prof. at UC-Berkeley]

Mapping is a "technical" term meaning the relationship between two things: between the controls, their movements, and the results in the world.

[The Design of Everyday Things by Don Norman]

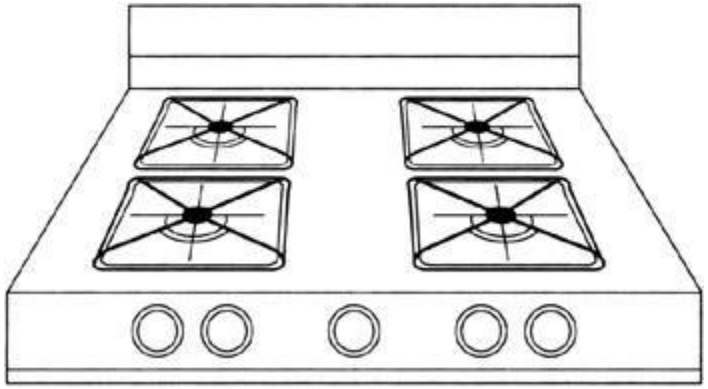
A “natural” mapping takes advantage of physical analogies and cultural norms, leading to an “immediate” understanding


Mapping

Mapping

Mapping


FM2
A.SEL
105.1
1:26
Premium Audio System
ANTI THEFT
DISP
SCAN
AM
FM
CD/AUX
A.SEL
CLOCK
4PC1
VOL
PUSH
PWR
1
H
DISC +
2
M
RPT
3
R
RDM
SEEK/SKIP
DISC –
4
5
6
TUNE
FOLDER
PUSH
SOUND
MP3 WMA
OFF
MAX
A/C
A/C



Mapping

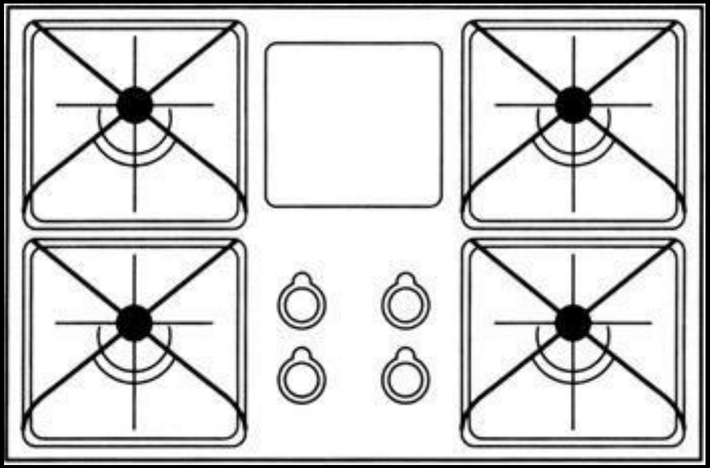
The ***result* of using the control** is reasonably clear: A burner will heat up when you turn a knob.

However, **the *target* of the control**—which burner will get warm—is unclear. Does twisting the left-most knob turn on the left-front burner, or does it turn on the left-rear burner?

Users must find out by trial and error or by referring to the tiny icons next to the knobs. The unnaturalness of the mapping compels users to figure this relationship out anew every time they use the stove .

[The Design of Everyday Things by Don Norman and About Face 2.0 by Cooper and Reimann]

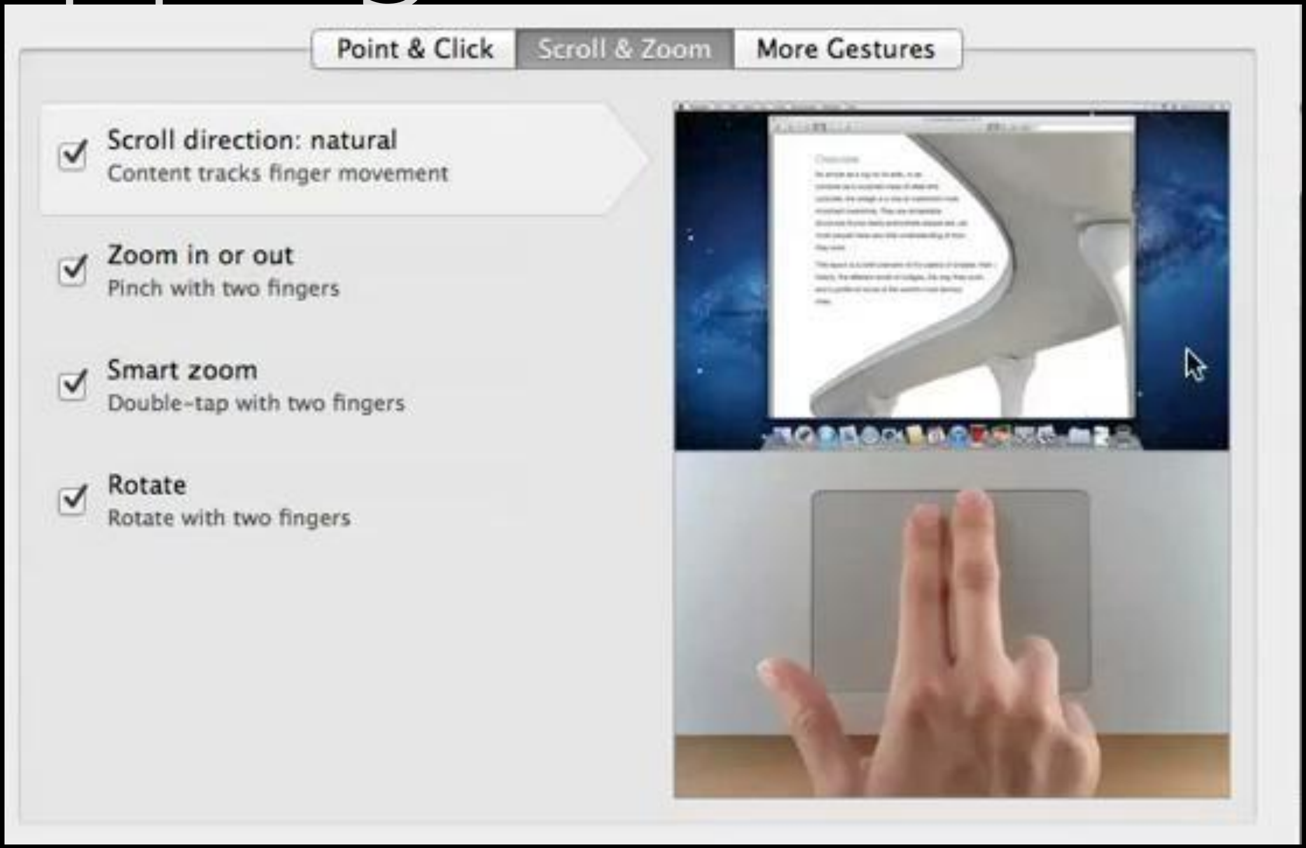
Mapping



[The Design of Everyday Things by Don Norman and About Face 2.0 by Cooper and Reimann]

Mapping

Mapping



[http://youtu.be/Gpa9_Fo0muM]

Feedback—sending back to the user information about what action has actually been done & what result has been accomplished—is a well known concept in the science of control and information theory.

Don Norman
Cognitive Scientist / Author

[The Design of Everyday Things by Don Norman]

Feedback

Effect improves
when immediate
and synchronized
with user action


Feedback


Use Exit
Feedback